

NPCL: Negative-Preserving Contrastive Learning under Noisy Correspondence

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Abstract. Cross-modal matching heavily relies on well-aligned image-text data to learn reliable semantic correspondences. However, web-collected datasets inevitably contain noisy correspondence, where mismatched image-text pairs are incorrectly treated as positives, leading to severe performance degradation. Existing approaches primarily focus on suppressing noisy samples through reweighting or filtering, but often overlook the critical role of negative samples, resulting in insufficient discriminative learning, especially under high-noise conditions. In this paper, we propose **Negative-Preserving Contrastive Learning** under Noisy Correspondence (NPCL). This robust cross-modal learning framework explicitly preserves and enhances negative sample supervision. NPCL first aggregates image features into multiple complementary views and employs a bidirectional cross-modal evidence extractor to model semantic consistency across modalities. Based on the extracted evidence, a Dirichlet distribution is used to jointly estimate matching belief and uncertainty, enabling reliable identification of noisy correspondences and differentiated optimization. Furthermore, NPCL introduces noise-aware and noise-decoupled contrastive losses that suppress misleading noisy positives while maintaining effective negative discrimination. Extensive experiments on MS-COCO, Flickr30K demonstrate that NPCL consistently achieves stable and superior retrieval performance across diverse noise levels, validating its effectiveness and robustness in noisy cross-modal matching scenarios.

Keywords: Cross-Modal Matching · Noisy Correspondence Learning · Trusted Robust Loss.

1 Introduction

Cross-modal matching has been widely studied in multimodal learning, where heterogeneous features are projected into a shared embedding space to maxi-

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mize the similarity of matched pairs and minimize that of mismatched pairs for semantic alignment [2–4]. Despite strong performance on benchmark datasets, these methods largely depend on large-scale, high-quality, and well-aligned cross-modal data [5].

Unlike objective image labels, textual descriptions are inherently subjective and diverse, making high-quality image–text annotation costly. As a result, large-scale datasets are typically collected from web-based image–text co-occurrences, which inevitably introduce noisy correspondence and significantly degrade cross-modal matching performance [6, 7].

Although noisy correspondence is underexplored in cross-modal matching, noise-label learning in classification has been widely studied through sample selection [5, 8, 9] and loss reweighting [10, 11]. However, these methods are not directly applicable, as noisy correspondence represents instance-level noise arising from pairwise relationships rather than explicit labels, resulting in more fine-grained decision boundaries.

Several studies address noisy correspondence, including NCR [12], DECL [13], and Bicro [14]. These methods rely on pseudo-label refinement, uncertainty modeling, or clean positive selection, implicitly assuming reliable positive pairs during training. However, under high-noise conditions, positive pair reliability deteriorates, posing significant challenges for conventional contrastive learning frameworks.

To mitigate noise, existing methods often apply uniform contrastive constraints to noisy samples, aiming to suppress their influence. Such strategies, however, tend to overlook the critical role of negative samples. Excessive suppression of noisy gradients may inadvertently weaken negative supervision, leading to insufficient negative optimization and degraded discriminative capability. Therefore, balancing noise suppression and negative sample preservation is crucial for robust and effective cross-modal matching.

To address this issue, we propose **Negative-Preserving Contrastive Learning** under Noisy Correspondence (NPCL). Instead of simply down-weighting noisy samples, NPCL rebalances the roles of noisy and negative samples through principled modeling. Specifically, image features are aggregated into multiple complementary views, and a bidirectional evidence extractor is employed to learn cross-modal evidence across views. These evidence parametersize a Dirichlet distribution, enabling the joint estimation of matching belief and overall uncertainty. Based on these estimates, samples are partitioned into clean and noisy sets, providing a reliable foundation for subsequent optimization.

On all samples, we introduce Noise-Aware Negative-Enhanced Contrastive Loss, which emphasizes negative separation while suppressing the influence of noisy positive pairs, thereby improving negative sample utilization. For noisy samples, we further apply Noise-Decoupled Contrastive Loss, which decouples the optimization of noisy positives and negatives, allowing negative samples to retain their discriminative contribution while penalizing noisy positives. Together, this strategy significantly enhances robustness and discrimination under noisy correspondence.

We evaluate NPCL on MS-COCO and Flickr30K with varying noise ratios. Experimental results demonstrate that NPCL achieves competitive or superior performance across diverse noise settings, validating its effectiveness and robustness.

The main contributions of this work are summarized as follows:

- We propose a robust cross-modal learning framework that explicitly preserves and enhances the role of negative samples to mitigate noisy correspondence.
- We leverage bidirectional cross-modal evidence and Dirichlet uncertainty modeling to reliably identify noisy correspondences and support differentiated optimization.
- Extensive experiments on synthetic and real noisy datasets demonstrate stable and superior retrieval performance under challenging noise conditions.

2 RELATED WORKS

Cross-Modal Matching. The goal of cross-modal matching is to bridge representation gaps across different modalities and establish semantic correspondences between heterogeneous data, such as images and text [8]. Existing methods mainly fall into two categories: embedding-based alignment and score-based matching.

Embedding-based approaches design modality-specific encoders to extract global features, which are projected into a shared embedding space for alignment and retrieval [15]. VSE [16] aligns image and text embeddings using ranking losses, while VSE++ [3] further introduces hard negative mining to improve discrimination. VSRN [7] and related works [17] employ graph convolutional networks (GCNs) to model intra-modal structure, enhancing semantic consistency. VSE ∞ [2] aggregates local features via learnable pooling to form global representations.

Score based methods focus on fine grained cross modal interactions by computing local similarities between fragments and aggregating them into an instance level matching score [22]. SCAN [11] aligns image regions with sentence words to capture local correspondences, while SGRAF [6] performs graph based reasoning over local similarities. NAAF [23] suppresses irrelevant fragments to improve robustness, and CHAN [24] filters mismatched alignments to reduce redundancy.

Despite strong benchmark performance, most existing methods assume accurately aligned training pairs. In practice, however, such assumptions often fail due to noisy data collection, making robust cross-modal matching under noisy correspondence an important research problem.

Learning with noisy Correspondence. Noisy Correspondence refers to semantic mismatches at the pairing level, where unmatched cross-modal samples

are incorrectly treated as positive pairs, which differs from noisy labels in classification that arise from incorrect category annotations. Among early efforts, Noisy Correspondence Rectify (NCR) [5] systematically addresses this problem by leveraging sample-level losses and a two-component Gaussian Mixture Model (GMM) [1] to separate clean and noisy samples, and refining soft correspondence labels using predicted similarities [25].

Subsequent works extend this paradigm in different ways. MSCN [13] trains a meta network on clean samples to generate more reliable soft correspondence estimates. BiCro [19] adopts a neighborhood consistency assumption to convert binary noisy labels into continuous soft labels and incorporates them as soft margins in triplet losses. DECL [20] improves robustness through noise-tolerant loss design, while UGNCL [26] utilizes Dirichlet-based uncertainty to partition data into multiple subsets and regularize triplet loss with uncertainty-aware soft margins.

In contrast to these methods, we tackle noisy correspondence from the perspective of negative sample optimization. Our approach enhances negative separation while suppressing interference from noisy positive pairs, thereby improving negative sample utilization. Specifically, NPCL constructs a Dirichlet distribution from multi-view bidirectional cross-modal evidence to estimate matching belief and uncertainty, and partitions samples into clean and noisy subsets. We then apply Noise-Aware Negative-Enhanced Contrastive Loss on all samples and Noise-Decoupled Contrastive Loss on noisy samples to decouple positive-negative optimization, leading to more robust and discriminative cross-modal alignment under high noise.

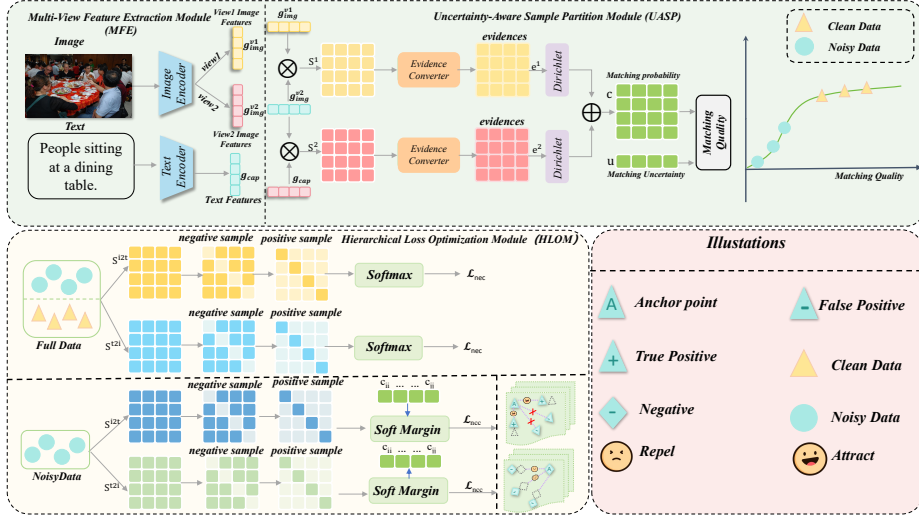


Fig. 1. Overview diagram of the model architecture.

3 METHODOLOGY

Multi-View Feature Extraction Module(MFE) To capture multi-view bidirectional semantics between images and text in a unified embedding space, we propose the Multi-View Dual-Channel Semantic Feature Extraction Module.

Given local image region features extracted by Fast R-CNN, $\{\mathbf{v}_r\}_{r=1}^R$ with $\mathbf{v}_r \in \mathbb{R}^{d_v}$, We employ two independent fully connected layers to obtain multi-view image embeddings:

$$\mathbf{E}_{\text{img}}^k = f_{\text{img}}^k(\{\mathbf{v}_r\}_{r=1}^R), \quad k \in \{1, 2\}, \quad (1)$$

where $\mathbf{E}_{\text{img}}^k \in \mathbb{R}^D$ and $D = 1024$. Each embedding is ℓ_2 -normalized as

$$\hat{\mathbf{E}}_{\text{img}}^k = \frac{\mathbf{E}_{\text{img}}^k}{\|\mathbf{E}_{\text{img}}^k\|_2}. \quad (2)$$

An initial global visual feature is computed by average pooling:

$$\mathbf{m}_{\text{img}} = \frac{1}{R} \sum_{r=1}^R \mathbf{v}_r, \quad (3)$$

The final image-level representation is obtained via a visual self-attention module:

$$\mathbf{g}_{\text{img}} = \text{VisualSA}(\hat{\mathbf{E}}_{\text{img}}, \mathbf{m}_{\text{img}}). \quad (4)$$

Given a sentence $\{w_t\}_{t=1}^T$, word embeddings $\{\mathbf{e}_t\}_{t=1}^T$ are encoded by a Bi-GRU to produce contextual features $\{\mathbf{h}_t\}_{t=1}^T$. The text embedding $\mathbf{E}_{\text{cap}} \in \mathbb{R}^{T \times d_h}$ is ℓ_2 -normalized to obtain $\hat{\mathbf{E}}_{\text{cap}}$. The initial global text feature is computed by average pooling:

$$\mathbf{m}_{\text{cap}} = \frac{1}{T} \sum_{t=1}^T \mathbf{h}_t, \quad (5)$$

The final text representation is produced by a textual self-attention module:

$$\mathbf{g}_{\text{cap}} = \text{TextSA}(\hat{\mathbf{E}}_{\text{cap}}, \mathbf{m}_{\text{cap}}) \quad (6)$$

Uncertainty-Aware Sample Partition Module (UASP) To dynamically identify variations in matching quality among sample pairs during training and enable precise differentiated treatment, thereby improving robustness under noisy data, we propose an Uncertainty-Aware Sample Partition Module. This module takes the dual-view visual semantic features $\mathbf{g}_{\text{img}}^{v1}$ and $\mathbf{g}_{\text{img}}^{v2}$ extracted by MFE, together with the textual semantic feature $\mathbf{g}_{\text{cap}}$, as inputs. It computes similarity matrices between visual and textual representations from two views, denoted as S^1 and S^2 , and performs uncertainty modeling via a Dirichlet distribution to estimate matching quality and partition samples accordingly.

Specifically, we first compute two similarity matrices $\mathbf{S}^1, \mathbf{S}^2 \in \mathbb{R}^{B \times B}$ as

$$\mathbf{S}^1 = \text{sim}(\mathbf{g}_{img}^{v1}, \mathbf{g}_{cap}), \quad \mathbf{S}^2 = \text{sim}(\mathbf{g}_{img}^{v2}, \mathbf{g}_{cap}), \quad (7)$$

where $\text{sim}(\cdot, \cdot)$ denotes the similarity function.

To transform similarities into non-negative matching evidence, we apply a Softplus function followed by exponential scaling:

$$\mathbf{e}^v = \exp\left(\frac{\text{Softplus}(\mathbf{S}^v)}{\tau}\right), \quad v \in \{1, 2\}, \quad (8)$$

where $\mathbf{e}^v$ denotes the matching evidence matrix for view v , τ is a temperature parameter controlling evidence scaling, and $\text{Softplus}(x) = \log(1 + e^x)$ ensures positivity.

Each view is then modeled with a Dirichlet distribution parameterized by $\boldsymbol{\alpha}^v = \mathbf{e}^v + \mathbf{1}$. To jointly leverage evidence from both views, we fuse the Dirichlet parameters using Dempster-Shafer evidence theory [21]:

$$\boldsymbol{\alpha}^{\text{fused}} = \boldsymbol{\alpha}^1 \oplus \boldsymbol{\alpha}^2, \quad (9)$$

where $\boldsymbol{\alpha}^{\text{fused}}$ denotes the fused evidence parameters and $\oplus$ represents the Dempster combination operator.

Based on the fused Dirichlet parameters, we compute the uncertainty and confidence for each sample:

$$u_i = \frac{K}{S_i}, \quad c_i = \frac{\alpha_i^{\text{fused}} - 1}{S_i}, \quad S_i = \sum_{k=1}^K \alpha_{i,k}^{\text{fused}}, \quad (10)$$

where u_i denotes the overall matching uncertainty of sample i , c_i represents the matching probability, α_i^{fused} is the fused evidence parameter of the matching class, S_i is the total evidence strength, and K is the number of samples in a batch.

To partition samples into clean and noisy subsets, we define a matching quality score as

$$m_i = \frac{1}{u_i} + c_i, \quad (11)$$

which jointly reflects overall uncertainty and matching confidence. Finally, based on m_i , we apply the clustering of k -means to divide all samples into a clean set S_C and a noisy set S_N .

Hierarchical Loss Optimization Module(HLOM) To enable efficient and robust training under noisy data and improve retrieval performance on noisy samples, we design a hierarchical loss optimization module. This module first computes a noise-robust contrastive loss to enhance negative sample separation and suppress interference from noisy positive pairs, thereby improving negative sample utilization. It then leverages the sample matching quality c_i produced by the UASP module to compute an asymmetric triplet loss, which decouples

Table 1. Performance comparison of multimodal retrieval models on Flickr30K and MS-COCO under different noise ratios.

Noise	Methods	Flickr30K							MS-COCO 1K						
		Image to Text			Text to Image				Image to Text			Text to Image			
		R@1	R@5	R@10	R@1	R@5	R@10	rSum	R@1	R@5	R@10	R@1	R@5	R@10	rSum
20%	SCAN	59.1	83.4	90.4	36.6	67.0	77.5	414.0	66.2	91.0	96.4	45.0	80.2	89.3	468.1
	VSRN	58.1	82.6	89.3	40.7	68.7	78.2	417.6	25.1	59.0	74.8	17.6	49.0	64.1	289.6
	IMRAM	63.0	86.0	91.3	41.4	71.2	80.5	433.1	68.6	92.8	97.6	52.7	85.0	91.0	490.7
	SGR	62.8	86.2	92.2	41.4	72.3	80.4	433.3	67.8	91.7	96.2	55.9	83.5	90.1	482.2
	SGR-C	72.8	90.8	95.4	56.4	82.1	88.6	486.1	75.4	95.2	97.9	60.1	88.5	94.8	511.9
	NCR	73.5	93.2	96.6	56.9	82.4	88.5	491.1	76.6	95.6	98.2	60.8	88.8	95.0	515.0
	DECL	77.5	93.8	97.0	56.1	81.8	88.5	494.7	78.5	95.9	98.4	61.7	89.3	95.4	518.2
	MSCN	77.4	94.9	97.6	59.6	83.2	89.2	501.9	77.1	97.2	98.8	64.3	<u>90.4</u>	<u>95.8</u>	524.6
	BiCro	78.1	94.4	97.5	60.4	<u>84.4</u>	<u>89.9</u>	504.7	78.8	96.1	98.6	63.7	90.3	95.7	523.2
	UGNCL	<u>78.4</u>	95.8	<u>97.8</u>	<u>59.8</u>	<u>84.3</u>	89.5	<u>505.6</u>	<u>79.5</u>	<u>97.2</u>	<u>99.0</u>	<u>63.7</u>	90.9	96.0	<u>526.3</u>
	NPCL	79.0	95.6	97.8	59.4	84.5	90.3	506.6	81.5	97.3	99.0	63.7	90.1	95.5	527.1
50%	SCAN	27.7	57.6	68.8	16.2	39.3	49.8	259.4	40.8	73.5	84.9	5.4	15.1	21.0	240.7
	VSRN	14.3	37.6	50.0	12.1	30.0	39.4	183.4	23.5	54.7	69.3	16.0	47.8	65.9	277.2
	IMRAM	9.1	26.6	38.2	2.7	8.4	12.7	97.7	21.3	60.2	75.9	22.3	52.8	64.3	296.8
	SGR	36.9	68.1	80.2	29.3	56.2	67.0	337.7	60.6	87.4	93.6	46.0	74.2	79.0	440.8
	SGR-C	69.8	90.3	94.8	50.1	77.5	85.2	467.7	71.7	94.1	97.7	57.0	86.6	93.7	500.8
	NCR	72.9	93.0	96.3	54.3	79.8	86.5	482.8	74.6	94.7	97.8	59.1	87.8	94.5	508.5
	DECL	72.7	92.0	95.8	54.8	80.4	<u>87.5</u>	483.2	76.1	95.0	98.3	60.5	88.7	94.9	513.5
	MSCN	<u>74.4</u>	93.2	96.0	55.3	80.4	<u>86.8</u>	486.1	<u>77.5</u>	96.0	<u>98.7</u>	60.7	89.1	94.9	517.1
	BiCro	73.1	91.4	96.1	53.7	80.2	87.4	481.9	76.2	96.2	98.4	61.6	89.2	<u>95.1</u>	516.5
	UGNCL	74.1	<u>93.0</u>	96.9	56.7	<u>80.9</u>	87.3	<u>488.9</u>	77.2	<u>96.4</u>	98.7	<u>61.6</u>	<u>89.7</u>	95.3	<u>518.9</u>
	NPCL	75.2	92.6	<u>96.7</u>	<u>56</u>	82.3	88.7	491.5	77.7	97	98.2	63.8	89.8	94.8	521.3
70%	SCAN	5.6	19.3	27.4	2.2	8.0	12.8	75.3	18.1	43.1	57.4	0.3	1.3	2.3	122.5
	VSRN	0.8	2.5	4.1	0.5	1.5	2.7	12.1	5.1	15.7	24.6	2.5	8.8	13.3	70.0
	IMRAM	1.3	3.1	3.9	0.3	1.2	2.8	12.6	7.1	20.0	33.4	5.3	15.2	22.0	103.0
	SGR*	17.9	42.1	51.9	14.6	31.0	40.8	198.3	35.7	71.2	85.4	51.6	65.8	79.0	368.7
	SGR-C	65.0	89.3	94.7	48.1	74.5	81.1	452.7	69.8	93.6	97.5	56.5	86.0	93.4	496.8
	NCR	16.1	38.5	52.8	11.0	29.5	41.4	189.3	35.4	69.5	83.4	31.5	66.4	81.1	367.3
	DECL	60.4	85.4	90.8	42.8	71.5	80.7	431.7	70.5	94.7	98.8	56.1	85.3	93.2	498.6
	MSCN	69.0	89.3	93.0	49.2	73.1	79.0	452.6	74.4	94.9	97.7	58.8	87.2	<u>93.7</u>	506.7
	BiCro	65.2	88.0	93.2	48.0	74.1	81.6	450.1	73.9	<u>95.5</u>	97.8	59.2	<u>87.8</u>	93.6	507.9
	UGNCL	<u>70.3</u>	91.1	96.0	<u>52.4</u>	<u>77.6</u>	<u>83.4</u>	<u>470.8</u>	<u>75.5</u>	95.9	<u>98.4</u>	<u>59.6</u>	87.5	92.6	<u>509.7</u>
	NPCL	71.5	<u>89.4</u>	<u>94.2</u>	53	79.6	86.4	474.1	76.3	94.7	98.3	60.3	88.2	94.5	512.3

the optimization paths of positive and negative samples. As a result, the model penalizes noisy positives while preserving the discriminative power of negative samples.

To suppress the influence of noisy positive pairs, we explicitly penalize the similarity of all non-matching pairs $(i, j \neq j^*(i))$ during training. We optimize similarities in both image-to-text (i2t) and text-to-image (t2i) directions to enhance cross-modal alignment robustness. The loss for each direction is defined as:

$$\mathcal{L}_{i2t} = -\frac{1}{N} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq j^*(i)}}^N \log\left(1 - \frac{\exp(S_{(i,j)})}{\sum_{k=1}^N \exp(S_{(i,k)})}\right), \quad (12)$$

where N is the batch size, $j^*(i)$ denotes the target image corresponding to text query i , and $S_{(i,j)}$ represents the similarity score between text i and image j .

Similarly, the text-to-image loss is given by:

$$\mathcal{L}_{t2i} = -\frac{1}{N} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq j^*(i)}}^N \log\left(1 - \frac{\exp(S_{(i,j)})}{\sum_{k=1}^N \exp(S_{(i,k)})}\right). \quad (13)$$

The final bidirectional contrastive loss is computed as:

$$\mathcal{L}_{\text{nec}} = \frac{\mathcal{L}_{i2t} + \mathcal{L}_{t2i}}{2}, \quad (14)$$

where $\mathcal{L}_{\text{nec}}$ denotes the noise-aware negative-enhanced contrastive loss.

For the noisy sample subset $\mathcal{S}_{\text{noise}}$, to decouple the optimization of positive and negative samples while preserving negative discrimination, we introduce an asymmetric triplet loss:

$$\mathcal{L}_{\text{ncc}} = \frac{1}{|\mathcal{S}_{\text{noise}}|} \sum_{i \in \mathcal{S}_{\text{noise}}} \log(1 + \exp(\lambda \mu_i^p(s_{ii} - m_p))) + \sum_{j \neq i} \exp(\lambda \mu_{ij}^n(s_{ij} - m_n)), \quad (15)$$

where $\mathcal{L}_{\text{ncc}}$ is the adaptive triplet loss, $\lambda > 0$ is a scaling factor, m_p is set to $1 + m_0$ and m_n is set to $-m_0$. m_0 is the base margin. Here, s_{ii} denotes the similarity between sample i and its positive pair, while s_{ij} denotes the similarity between sample i and negative sample j .

To dynamically adjust supervision strength based on sample confidence, we introduce an adaptive margin mechanism:

$$\mu_i^p = \max(0, \sigma_i U_p - s_{ii}), \quad (16)$$

$$\mu_{ij}^n = \max(0, s_{ij} - U_n), \quad (17)$$

Where μ_i^p and μ_{ij}^n denote the adaptive margins for positive and negative samples, respectively. Here, $U_p = 1 + m_p$ is the ideal upper bound for positive similarity, and $U_n = -m_n$ is the ideal lower bound for negative similarity.

The confidence scaling factor is defined as:

$$\sigma_i = \frac{z^{c_i-1}}{z-1}, \quad (18)$$

Where $c_i \in [0, 1]$ denotes the matching probability of sample i , and $z > 1$ is a hyperparameter controlling the steepness of the scaling curve.

The overall training objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{nec}} + \alpha \mathcal{L}_{\text{ncc}}, \quad (19)$$

where $\mathcal{L}_{\text{total}}$ denotes the total loss and α is a dynamically adjusted weighting coefficient.

4 EXPERIMENTS

4.1 Experimental Settings

Our proposed NPCL can be readily integrated into existing cross-modal vision-language models and significantly improves robustness against noisy correspondence. For fair comparison, we follow the experimental settings of NCR except for NPCL specific parameters, and evaluate our method using SGR, SAF, and SGRAF as backbone networks. We conduct experiments under noise ratios $r = 0.2, 0.5$, and 0.7 . All experiments are conducted without additional data preprocessing or external data sources.

The embedding dimension D is set to 1024, and the mini batch size is 128. The loss weight α is set to 0.3, and the base margin m_0 is set to 0.2. For MS-COCO, models are trained for 20 epochs after a 5 epochs warm up, while for other datasets, training lasts 30 epochs following a 5 epochs warm up. During inference, similarities from different views are averaged for retrieval evaluation.

4.2 Performance Comparison

Table 1 reports performance comparisons on Flickr30K and MS-COCO 1K under different noise ratios ($r \in \{0.2, 0.5, 0.7\}$). As the noise level increases, all methods suffer performance degradation to varying degrees. Nevertheless, NPCL consistently achieves the best or near-best performance across all noise settings and both datasets, demonstrating strong robustness against noisy correspondence.

Under moderate noise ($r = 0.2$), NPCL already shows consistent advantages. It achieves the highest rSum of 506.6 on Flickr30K and 527.1 on MS-COCO, outperforming strong baselines such as UGNCL and BiCro. When the noise ratio increases to $r = 0.5$, the performance gap becomes more evident. NPCL reaches an rSum of 491.5 on Flickr30K and 521.3 on MS-COCO, exceeding UGNCL by a clear margin. This indicates that preserving negative discrimination while suppressing noisy positives is critical when noisy and clean samples are comparable in scale.

In the extreme noisy setting ($r = 0.7$), most methods experience severe performance collapse, whereas NPCL remains notably stable. On Flickr30K, NPCL achieves an rSum of 474.1, surpassing UGNCL (470.8), and attains a clear advantage on the T2I task with an R@10 of 86.4. Similar trends are observed on MS-COCO, where NPCL obtains the highest rSum of 512.3. These results confirm that NPCL effectively mitigates the adverse impact of noisy correspondence and maintains robust cross-modal discrimination under challenging noise conditions.

4.3 Ablation Study

To investigate the contribution of each key component in NPCL, we conduct ablation experiments on Flickr30K and MS-COCO 1K under a moderate noise ratio ($r = 0.2$). The results are summarized in Table 2. We analyze the effects of different components as follows.

Table 2. Ablation study results (rSum) on Flickr30K and MS-COCO

Noise	Methods	Flickr30K rSum	MS-COCO 1K rSum
0.2	w/o K-Means	496.1	526.58
	w/o $\mathcal{L}_{ncc}$	485.7	516.18
	w/o UASP	496.5	516.98

- **w/o K-Means.** Removing K-Means clustering leads to a clear performance drop on both datasets, with rSum decreasing to 496.1 on Flickr30K and 526.58 on MS-COCO. K-Means clustering is used to partition samples based on matching quality, enabling effective separation between noisy and relatively clean pairs. Without this partitioning mechanism, noisy samples are mixed with clean ones during training, causing noisy correspondences to continuously interfere with model optimization. This degrades negative sample separation and weakens the discriminative decision boundary, highlighting the importance of K-Means for noise-aware training.
- **w/o $\mathcal{L}_{ncc}$.** Removing the noise-decoupled contrastive loss $\mathcal{L}_{ncc}$ results in a noticeable degradation of rSum, dropping to 485.7 on Flickr30K and 516.18 on MS-COCO. $\mathcal{L}_{ncc}$ introduces confidence-aware asymmetric triplet constraints on noisy samples, explicitly decoupling the optimization paths of noisy positives and negatives. Without this loss, noisy positive pairs suppress negative gradients, leading to insufficient negative optimization. This result demonstrates that $\mathcal{L}_{ncc}$ plays a critical role in preserving negative discriminability under noisy correspondence.
- **w/o UASP.** Removing the uncertainty-aware sample partition module also causes performance degradation, with rSum decreasing to 496.5 on Flickr30K and 516.98 on MS-COCO. UASP models match uncertainty via Dirichlet distributions and provide reliable matching quality estimates for sample partitioning. Without uncertainty-aware guidance, samples with different confidence levels are treated uniformly, reducing the effectiveness of noise suppression and differentiated optimization. This confirms the necessity of uncertainty modeling for robust training under noisy correspondence.

4.4 Parametric Analysis

We first analyze the effect of the base margin m_0 on cross-modal retrieval performance under a moderate noise ratio ($r = 0.2$). On both MS-COCO and Flickr30K, the performance exhibits a clear bell-shaped trend as m_0 varies. Specifically, increasing m_0 from 0.1 to 0.3 consistently improves rSum on both datasets, with the best performance achieved at $m_0 = 0.3$. This indicates that a moderate margin effectively enlarges the separation between positive and negative pairs, enhancing discriminative learning under noisy correspondence. However, further increasing m_0 leads to noticeable performance degradation, as

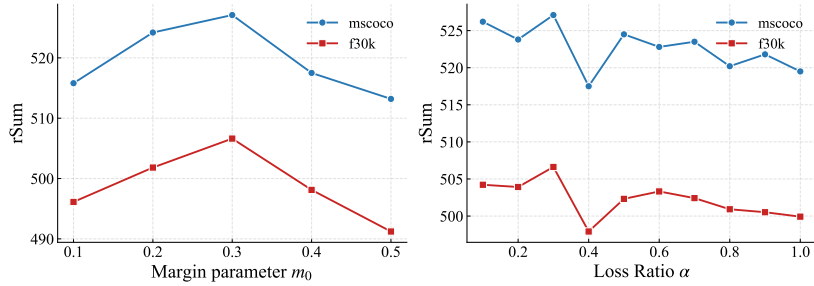


Fig. 2. Impact of m_0 and α on retrieval performance.

overly large margins impose excessively strict constraints on positive pairs, which is detrimental when noisy positives are present.

We further conduct a sensitivity analysis on the loss balancing parameter α . Overall, the performance on both datasets remains relatively stable across a wide range of α , while showing non-monotonic variations. The optimal performance is consistently observed around $\alpha = 0.3$ on both MS-COCO and Flickr30K. When α is too small, the noise-aware loss has limited influence, resulting in insufficient suppression of noisy correspondence. Conversely, overly large α values cause the optimization to overemphasize noise handling, which may suppress useful gradients from reliable samples and degrade discriminative capability.

4.5 Case Study

Fig. 3 compares retrieval results of the full NPCL model and its variant without $\mathcal{L}_{ncc}$ on the same example. The full model correctly ranks the target image/text at the top position in both T2I and I2T tasks, while removing $\mathcal{L}_{ncc}$ leads to mis-ranking of visually or semantically similar distractors. This comparison highlights that $\mathcal{L}_{ncc}$ effectively suppresses noisy positives while preserving negative discriminability, resulting in more accurate and robust retrieval.



Fig. 3. Comparison of retrieval performance between the full version model and w/o $\mathcal{L}_{ncc}$ on the same case

Fig. 4 visualizes the sample partition results produced by the UASP module, where samples are divided into clean and noisy subsets. The value shown in the upper-left corner denotes the *Matching Quality*. Samples with higher Matching Quality are identified as clean samples, while those with lower values are classified as noisy samples.



Fig. 4. Noise Partitioning Example

5 Conclusion

This work addresses noisy correspondence in cross-modal matching by highlighting the often-overlooked role of negative samples. We propose Negative-Preserving Contrastive Learning (NPCL), which rebalances noisy positives and informative negatives via multi view cross-modal evidence modeling and Dirichlet based uncertainty estimation. NPCL further introduces noise aware and noise decoupled contrastive losses to suppress misleading supervision while preserving discriminative learning from negative samples. Extensive experiments on MS-COCO, Flickr30K, and Concept Captions demonstrate consistent gains across diverse noise levels. These results validate NPCL as an effective and robust framework for cross-modal retrieval under noisy correspondence.

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